

Koijam Goddessia Konstatina

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Technology Lead | Software Engineer | AI/ML Researcher

PROFESSIONAL SUMMARY

Technology Professional with **9+ years of experience** currently serving as Technology Lead with prior experience as Software Engineer, Network/Application Administrator and Technical Support.

Strong expertise in **software engineering, application development, system architecture, and technical leadership**, with experience delivering scalable enterprise and government applications. Proven skills in **system design, performance optimization, requirements analysis, code reviews, mentoring, team leadership, and end-to-end SDLC delivery**.

Currently pursuing a PhD in Computer Science & Engineering(Part-Time), with research focused on OCR, computer vision, deep learning, and character recognition. Brings a combination of software engineering leadership and applied AI/ML research expertise.

CORE SKILLS

Languages: C#, Python, JavaScript, PHP, Java, SQL

Backend & Frameworks: .NET, ASP.NET Core, Web API, REST APIs, Node.js

Frontend: React.js, next.js, JavaScript

Architecture: Microservices, System Design, Software Architecture, API Design, Design Patterns

Databases: SQL Server, MySQL, MongoDB

Cloud & DevOps: AWS, Azure, Docker, Git/GitHub, GitHub Actions, CI/CD

AI/ML: LLMs, Generative AI, Ollama, OpenAI API, PyTorch, OpenCV, OCR, Deep Learning, Transfer Learning

PROFESSIONAL EXPERIENCE

Technology Lead | Knowext Inc., Montreal, Canada - Remote

September 2023 – Present

- Lead software engineering activities across application development, technical design, system enhancement, and production support.
- Design backend services and APIs using **C#, .NET, ASP.NET Core, SQL Server, and cloud technologies**.
- Contribute to **system architecture, microservices, API design, scalability, and performance optimization**.
- Conduct code reviews, mentor developers, and promote engineering standards and maintainable development practices.
- Gather requirements and collaborate with stakeholders throughout the **SDLC**, including design, development, deployment, and production support.
- Support automated deployment workflows using **GitHub Actions and CI/CD**.

Network Administrator | CDAC – RIMS Imphal

November 2020 – June 2023

- Managed infrastructure and application environments for a statewide **Learning Management System** supporting medical education.
- Performed monitoring, troubleshooting, system maintenance, access control, and technical support.
- Supported application availability, security, and data integrity across the platform.

Software Engineer | Cubeten (Govt. Empanelled)

July 2019 – October 2020

- Developed and maintained state-level digital platforms using **.NET, PHP, JavaScript, MySQL, and Web APIs**.
- Contributed to the **ILP Manipur Online Portal** and **Tengbang COVID-19 Support Portal**, including backend development, APIs, database operations, and application workflows.
- Optimized SQL queries, API logic, and data workflows and supported end-to-end SDLC activities.

Associate Software Developer | Mobimp

January 2019 – June 2019

- Developed application features, integrated APIs, improved UI responsiveness, and resolved software defects.
- Participated in Agile sprints and delivered iterative production improvements.

Technical Support / Web Application Maintainer | BuyBuy Manipur (Part Time)

August 2016 – January 2019

- Maintained web application content, inventory, pricing, and operational workflows.
- Performed application monitoring, troubleshooting, and backend maintenance.

KEY PROJECTS

AI ENGINEERING & GENERATIVE AI PROJECTS

Python | FastAPI | React/Next.js | LLMs | Ollama | OpenAI API | REST APIs | Git/GitHub

- Built end-to-end AI applications integrating **LLMs with modern software architectures**, including FastAPI backends and React/Next.js frontends.
- Integrated local LLMs using **Ollama** and explored hosted LLM/API-based integration.
- Applied modular architecture, API design, frontend/backend integration, and version-controlled development to AI applications.

NLUP ENGINE — KNOWLEDGE EXTRACTION SYSTEM

Document Processing | Regular Expressions | Rule-Based Extraction | SQL

- Developed a document-processing workflow for converting **unstructured documents in multiple formats into structured data**.
- Implemented rule-based information extraction using **regular expressions**, followed by structured database storage and retrieval.
- Designed an AI/NLP enhancement approach combining deterministic rules with AI for document classification, semantic extraction, entity recognition, normalization, and validation.

Campusoft — School Management System

.NET | SQL Server | JavaScript | Web APIs

- Contributed to a multi-module platform supporting **30+ educational institutions**, including feature development, performance, usability, and maintenance.

ILP Manipur Online Portal

.NET | Web APIs | SQL Server | JavaScript

- Developed and enhanced functionality for a government online permit platform.
- Contributed to backend development, API integration, database operations, and digital permit workflows.
- Supported deployment and production maintenance of the platform.

Tengbang COVID-19 Support Portal

.NET | Web APIs | SQL Server | JavaScript

- Contributed to the rapid development and delivery of a statewide public assistance platform during the COVID-19 pandemic.
- Developed application functionality, APIs, database operations, and integration components.

Manipur Education College Management Software

.NET | SQL Server | JavaScript

- Delivered feature enhancements and application improvements for education management workflows.
- Improved application usability and supported ongoing application maintenance and development.

RESEARCH & PUBLICATIONS

PHD RESEARCH(Ongoing) — COMPUTER VISION, OCR & DEEP LEARNING(Ongoing)

Title : Development of a Unified Framework for the Recognition of Printed and Handwritten Manipuri Script using OCR and Deep Learning Techniques.

Manipur Institute of Technology, Manipur University | 2024 – Present

- Conducting research on a unified framework for recognition of printed and handwritten Manipuri script using OCR and deep learning techniques.
- Developed image preprocessing, dataset preparation, and augmentation workflows for handwritten character recognition.
- Implemented transfer learning using **PyTorch** and evaluated AlexNet, ResNet-50, MobileNetV3, and EfficientNetV2.
- Research areas include **computer vision, OCR, image classification, deep learning, transfer learning, and pattern recognition.**

PUBLICATIONS:

“A Transfer Learning Approach for the Recognition of Handwritten Meetei/Meitei Mayek Digits”

Conference Paper | 2025

“Recognition of Handwritten Meetei/Meitei Mayek Digits using Transfer Learning Approach”

Journal Manuscript | 2026

“Performance Evaluation of Transfer Learning Architectures for Handwritten Meetei/Meitei Mayek Digit Recognition”

Journal Manuscript | 2026

EDUCATION

PhD in Computer Science & Engineering (Pursuing Part-Time)

Manipur Institute of Technology, Manipur University — 2024 – Present

M.Tech in Computer Science & Engineering

Manipur Institute of Technology, Manipur University — 2019 (89%)

B.E. in Computer Science & Engineering

Shridevi Institute of Engineering & Technology, VTU Belgaum — 2016

CERTIFICATIONS & TRAINING

- National Conference on Emerging Trends in Engineering Science & Management
- Workshop on Developments and Potential Applications of Superconductors
- Webinar Lead: “Supplementing Medical Education with E-Technology”

ACHIEVEMENTS

- Successfully deployed ILP Manipur Online Platform at the Chief Minister’s Secretariat.
- Contributed to Tengbang COVID-19 platform used statewide during the pandemic.
- Contributed to the development of Campusoft, supporting 30+ educational institutions.
- Delivered webinar sessions promoting e-learning adoption and technical literacy.